

HTML — Detailed Notes

English Guide for Web Development • Ahsan Web Studio

HTML stands for HyperText Markup Language. It is the standard markup language used to create the structure and content of web pages.

1. What is HTML?

Definition

HTML creates the structure of a webpage. It tells the browser what is a heading, paragraph, image, link, button, section and other content.

Important point

HTML is a markup language, not a programming language.

2. Basic HTML Document Structure

Example

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>My Website</title>
</head>
<body>
<h1>Hello World</h1>
</body>
</html>
```

DOCTYPE

Tells the browser that the document uses modern HTML5.

html

The root element of the HTML document.

head

Contains metadata, title, favicon, CSS references and other information about the page.

body

Contains the visible webpage content.

3. HTML Elements and Tags

Example

```
<p>Hello</p>
```

Opening tag

```
<p>
```

Content

Hello

Closing tag

```
</p>
```

Meaning

Together they form a paragraph element.

4. Headings

Tags

```
<h1> through <h6>
```

Example

```
<h1>Main Heading</h1>  
<h2>Section Heading</h2>  
<h3>Sub Heading</h3>
```

Tip

Use h1 for the main page heading and use h2/h3 to organize sections logically.

5. Paragraphs

Tag

```
<p>
```

Example

```
<p>I create clean and modern websites.</p>
```

Tip

Use separate paragraph elements for separate ideas.

6. Links

Tag

```
<a>
```

Example

```
<a href="portfolio.html">Portfolio</a>
```

href

The href attribute tells the browser where the link should go.

External link

```
<a href="https://www.google.com">Google</a>
```

7. Images

Tag

```
<img>
```

Example

```

```

src

Specifies the image file or URL.

alt

Provides alternative text describing the image. It is useful for accessibility and when an image cannot load.

8. Lists

Unordered list

```
<ul>
<li>HTML</li>
<li>CSS</li>
<li>JavaScript</li>
</ul>
```

Ordered list

```
<ol>
<li>Learn HTML</li>
<li>Learn CSS</li>
<li>Learn JavaScript</li>
</ol>
```

li

Creates an individual list item.

9. div Element

Purpose

The div element groups content and is commonly used as a container for styling and layout.

Example

```
<div class="card">
<h2>Business Website</h2>
<p>Modern websites for businesses.</p>
</div>
```

10. Classes

Purpose

A class gives one or more elements a reusable name that CSS or JavaScript can target.

Example

```
<div class="card">Content</div>
```

CSS example

```
.card {
padding: 20px;
```

```
border-radius: 15px;
}
```

11. IDs

Purpose

An id identifies a particular element on a page.

Example

```
<section id="services">...</section>
```

Anchor link

```
<a href="#services">Services</a>
```

12. Forms

Example

```
<form>
<label>Name</label>
<input type="text" placeholder="Enter your name">

<label>Email</label>
<input type="email" placeholder="Enter your email">

<button type="submit">Submit</button>
</form>
```

Common input types

text, email, password, number, date, checkbox and radio.

13. Semantic HTML

Meaning

Semantic elements describe the purpose of content and make page structure clearer.

Common elements

```
<header>, <nav>, <main>, <section>, <article>, <footer>
```

Example

```
<header>...</header>
<nav>...</nav>
<main>
<section>...</section>
</main>
<footer>...</footer>
```

14. HTML Attributes

Meaning

Attributes provide additional information about an HTML element.

Common attributes

class, id, href, src, alt, title, style, name, value, type and placeholder.

Example

```

```

15. HTML Comments

Example

```
<!-- This is the Services section -->
```

Purpose

Comments are notes for developers. They are not displayed as normal webpage content.

16. HTML + CSS + JavaScript

HTML

Creates the structure.

CSS

Controls the visual design and layout.

JavaScript

Adds behavior and interaction.

Easy memory trick

HTML = Structure, CSS = Design, JavaScript = Interaction.

17. A Small Complete Website

Example

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport"
content="width=device-width, initial-scale=1.0">
<title>Ahsan Web Studio</title>
</head>
<body>

<header>
<h1>Ahsan Web Studio</h1>
<nav>
<a href="index.html">Home</a>
<a href="portfolio.html">Portfolio</a>
</nav>
</header>

<main>
<section>
<h2>Welcome</h2>
<p>I create modern websites.</p>

</section>
</main>

<footer>
© 2026 Ahsan Web Studio
</footer>

</body>
</html>
```

18. Best Learning Order

Level 1

Document structure → headings → paragraphs → links → images.

Level 2

Lists → tables → forms → buttons → attributes.

Level 3

div → classes → IDs → semantic HTML.

Level 4

HTML + CSS → responsive design.

Level 5

HTML + CSS + JavaScript → interactive websites.

Level 6

GitHub → GitHub Pages → SEO → real projects.

HTML Practice Questions

1. What does HTML stand for?
2. What is the purpose of the <head> element?
3. What is the difference between an element and an attribute?
4. What is the purpose of the href attribute?
5. Why is the alt attribute useful for images?
6. What is the difference between class and id?
7. What are semantic HTML elements?
8. What is the role of the <body> element?
9. What is the difference between and ?
10. What are the roles of HTML, CSS and JavaScript?

Keep practicing by modifying your Ahsan Web Studio website and building small pages.